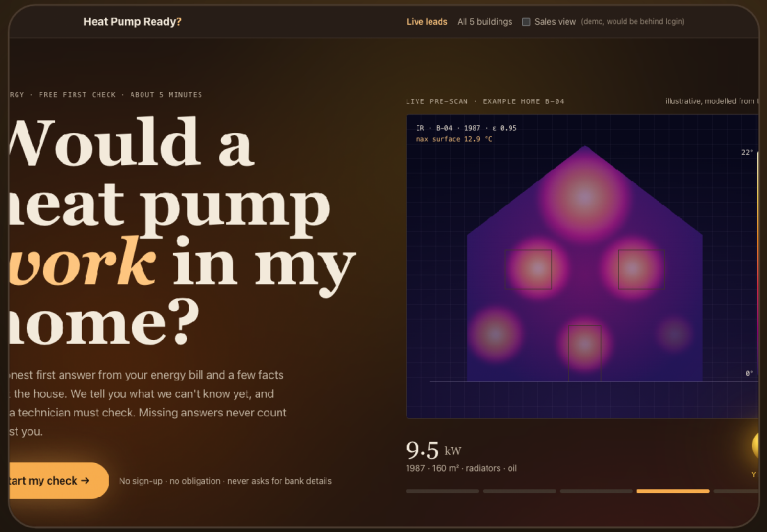
WATT THE HACK · MUNICH · 24 OCT 2026 · NUUENERGY CHALLENGE

● B-04 · 1987 · 160 m² · oil · the owner asks

“Would a heat pump *work* in my home?”

Heat Pump Ready? An honest first answer for homeowners, by voice or app, and a smarter visit decision for nuuEnergy’s sales team.



TEAM NOIDEA

TRACK: BUILDINGS & EVERYDAY ENERGY

01 · THE PROBLEM

Millions of homes still burn gas or oil. Most owners never get past *the first question.*

HOMEOWNER

“Would it even work here?”

Many wrongly assume they need floor heating or a full renovation.

TODAY

Only a visit answers it

A heating master spends about two hours on site to calculate the heat load.

NUUENERGY

Costly free visits

Some end in “not without major changes”; some homeowners never buy.

THE ASK

An honest first answer in under three minutes from the energy bill, building facts and a few phone photos.

02 · OUR ANSWER

We don’t just tell homeowners if they’re ready. We tell nuuEnergy *which visits are worth paying for.*

1 · Homeowner

Voice call, app or web

One question at a time, 8 European languages, photo link by SMS.

2 · Orchestrator

Astra + a guard

Blocks invented numbers or verdicts.

3 · Engine

Deterministic rules

Every number and verdict, from nuuEnergy’s guardrails.

4 · Sales desk

Live leads + briefing

With consent: next step, map, a page for the heating master.

AI talks. *Rules decide.*

One engine behind web app, phone app, voice agent and sales dashboard.
27 tools, also exposed over MCP.

03 · CALIBRATED & HONEST BY DESIGN

Three reference houses, *three matching verdicts.*

HOUSE	OURS	REFERENCE	OFF	VERDICT
B-01	9.8 kW	9.6 kW	+2%	Green
B-02	9.9 kW	9.7 kW	+2%	Green
B-03	16.6 kW	14.5 kW	+14%	Yellow

Two heat-load methods (floor area × age band, and the real bill) combined by a few named rules. No machine learning, no hidden constants: same input, same answer.

Missing data is never a “no”

Unknowns lead to yellow and a technician check, never red.

Age never rules a home out

Fraunhofer ISE: no correlation between build year and efficiency.

55 °C is a guide, not a wall

Verbraucherzentrale’s rule of thumb; well-sized radiators can work.

Single-pipe = a closer look

A strong flag with ways forward, as nuuEnergy’s own site describes.

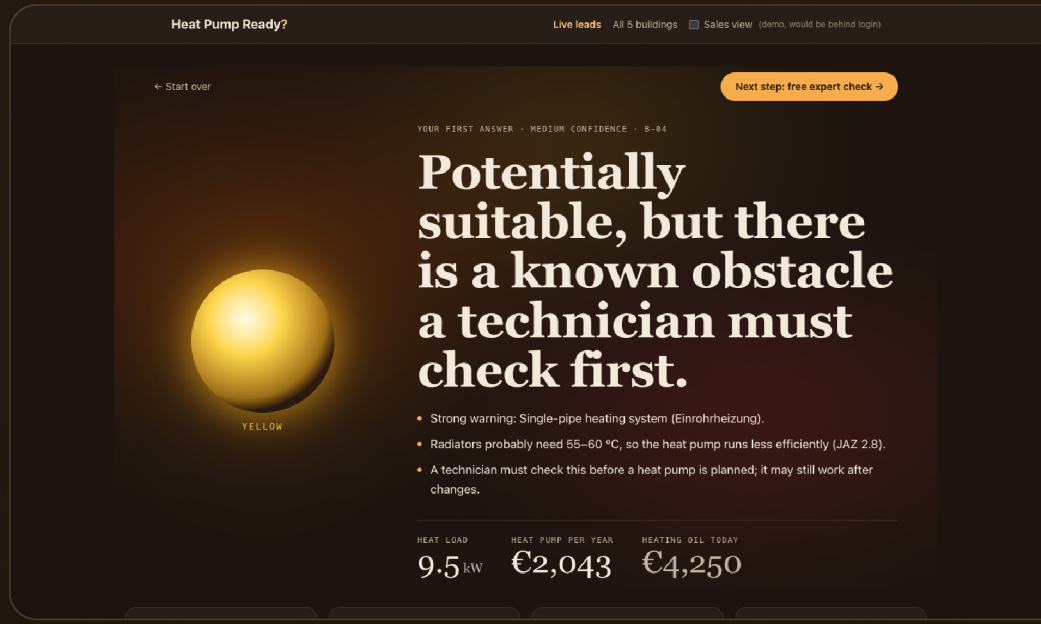
Calibrated on 3 labelled houses only · 13 public sources mapped rule by rule

04 · WHAT B-04'S OWNER SEES

Every result *explains itself.*

- **Heat demand** · area, insulation, real use
- **Heating-system fit** · flow temp, pipes, comfort
- **Installation feasibility** · space, electrics, permits
- **Information confidence** · what's known, what's open

Each with a status, the reason and the next check: always something to do, never a dead end.



05 · THE JURY'S HIDDEN HOUSES

Yellow isn't the end. *It's a to-do list.*

B-04 · 1987 · 160 m² · oil — 9.5 kW

Yellow: confirmed single-pipe heating, a closer look, not a no.

Heat pump €2,043 a year vs oil €4,250

B-05 · 2006 · 125 m² · gas — 10.4 kW

Yellow: the data contradicts itself: 216 kWh/m² for a 2006 house, 14 rooms in 125 m².

Heat pump €1,765–2,206 a year vs gas €2,970

PATH TO GREEN · B-04

TODAY	CHECK	CHECK	BIGGER WORK
Yellow, medium	55 °C test passes	Radiators sized	Loop fixed → green, high

The engine re-runs with each check assumed positive, cheapest first. B-01 reaches green with checks alone; B-03 and B-05 honestly stay yellow until insulation lowers their 83 W/m². A simulation, not a promise.

06 · A VOICE FOR OLDER HOMEOWNERS

Just call. AI reads the photos. *It never sets the verdict.*

Patient by design

Slower calm voice, numbers read back, “I don’t know” always fine, a human on request.

Eight languages

DE, EN, FR, IT, ES, NL, PL, PT, switching mid-call.

Asks only what matters

~9 core questions; more only if the answer would change the result.

2 / 2
live test calls correct

0
times the guard stepped in

Never asked
age · income · bank

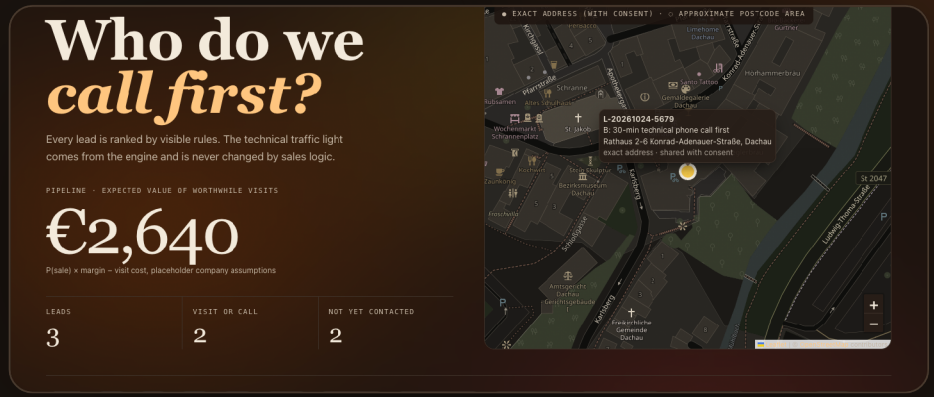
HELPER	MEASURED	SAFETY RULE
Radiator photo	4 of 5 right, 0 confidently wrong	Below 70% → “unclear”
Bill reader	Real SWM bill: 12,945 kWh, €765.02 correct	Caller confirms first
Meter cabinet, outdoor spot	2 of 4 exact, 0 confidently wrong	A technician confirms
Plain-language summary	5 of 5 kept every number and verdict	New numbers → rejected

Three independent looks per photo, then a vote. Disagreement → “a technician checks.” Small samples: signs, not accuracy figures.

07 · THE SALES DESK

Should we send *an expert?*

Live leads from voice and app, ranked, on a map. Click a lead and the map flies to the house; the drawer is a printable briefing for the heating master.



A
Book the on-site visit
~2-hour heat-load visit

B
30-min call first
Existing technical planning call

C
Ask for info first
Contradictions, missing photos

D
Not now
Info and subsidy guide

`EV = P(sale) × margin − visit cost` · inputs nuuEnergy sets live · ranked on building facts and answers, never age, income or personal traits.

08 · FAIR, PRIVATE, BUILT IN ONE DAY

Trust is *the product.*

Never asked, never scored

Age, income, health, nationality, bank details. Income bonus is calculated on-device only.

Consent before anything moves

Homeowner confirms, then shares; right to object to profiling stated.

Locked-down sales data

Leads, photos and addresses only behind a sales token; no keys in the app.

Scam-safe for older callers

Every call opens: nuuEnergy will never ask for bank details or passwords.

155
automated tests passing

27
tools for the assistant

8
languages by voice

3
front ends, one engine

Morning: engine & web app → Midday: sales tiers, subsidies → Afternoon: agent, MCP, voice → Evening: photo AI, live leads, phone app

Learn from real outcomes

Compare visits with predictions, tune the rules. Never a black box.

Faster voice turns

Today ~5–8 s per turn; streaming and caching bring it down.

Bigger photo test sets

nuuEnergy's labelled photos turn early signs into measured accuracy.

Into their CRM

JSON, e-mail or webhook today; the promises are settings, not code.

Fewer wasted visits.
More homes that say yes.

THANK YOU